

DASC7606

Deep Learning

Preliminary

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2025/26

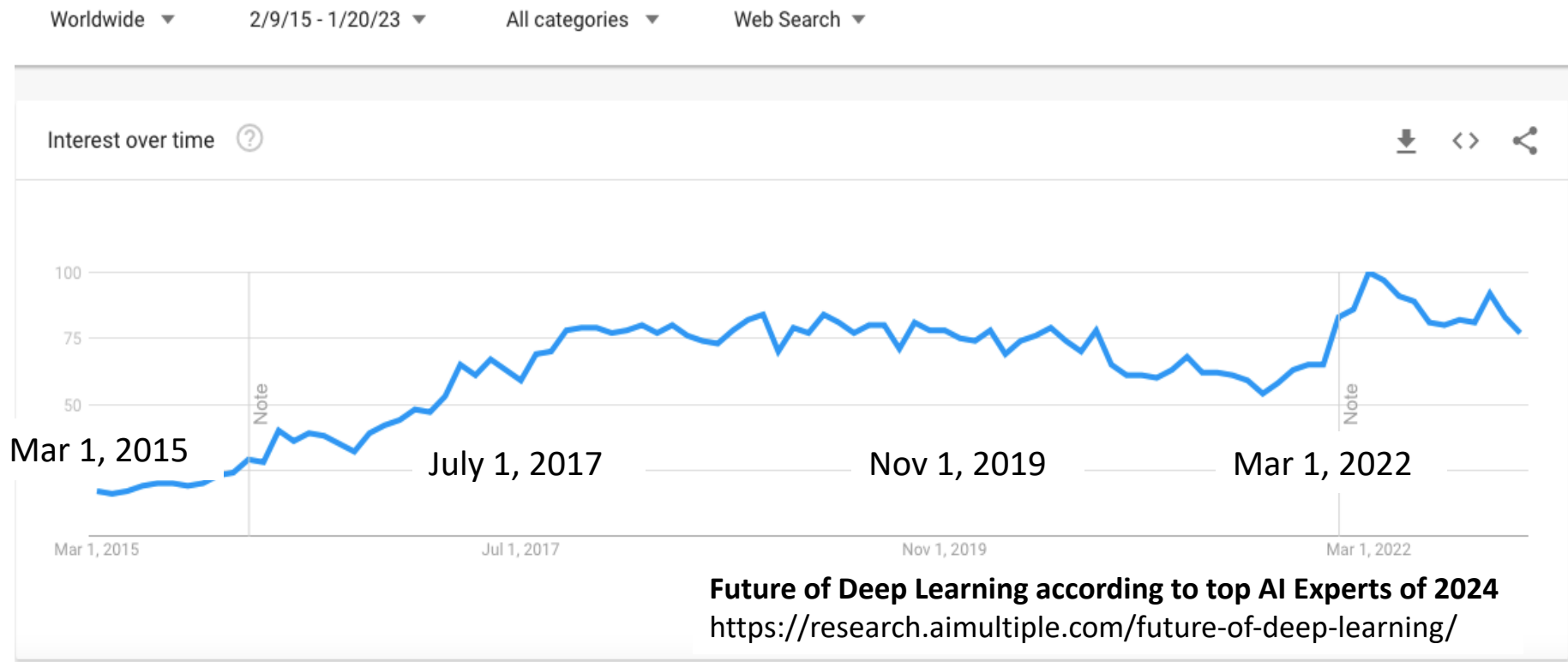
Why this Deep Learning course?

Ask the following questions

- Does the interest in AI actually increase in recent years?
- What kinds of AI jobs are trending?
- What are the most in-demand skills for these jobs?

How to measure the general interest in “Deep Learning”?

No. of searches about “Deep Learning” on Google indicates its popularity.

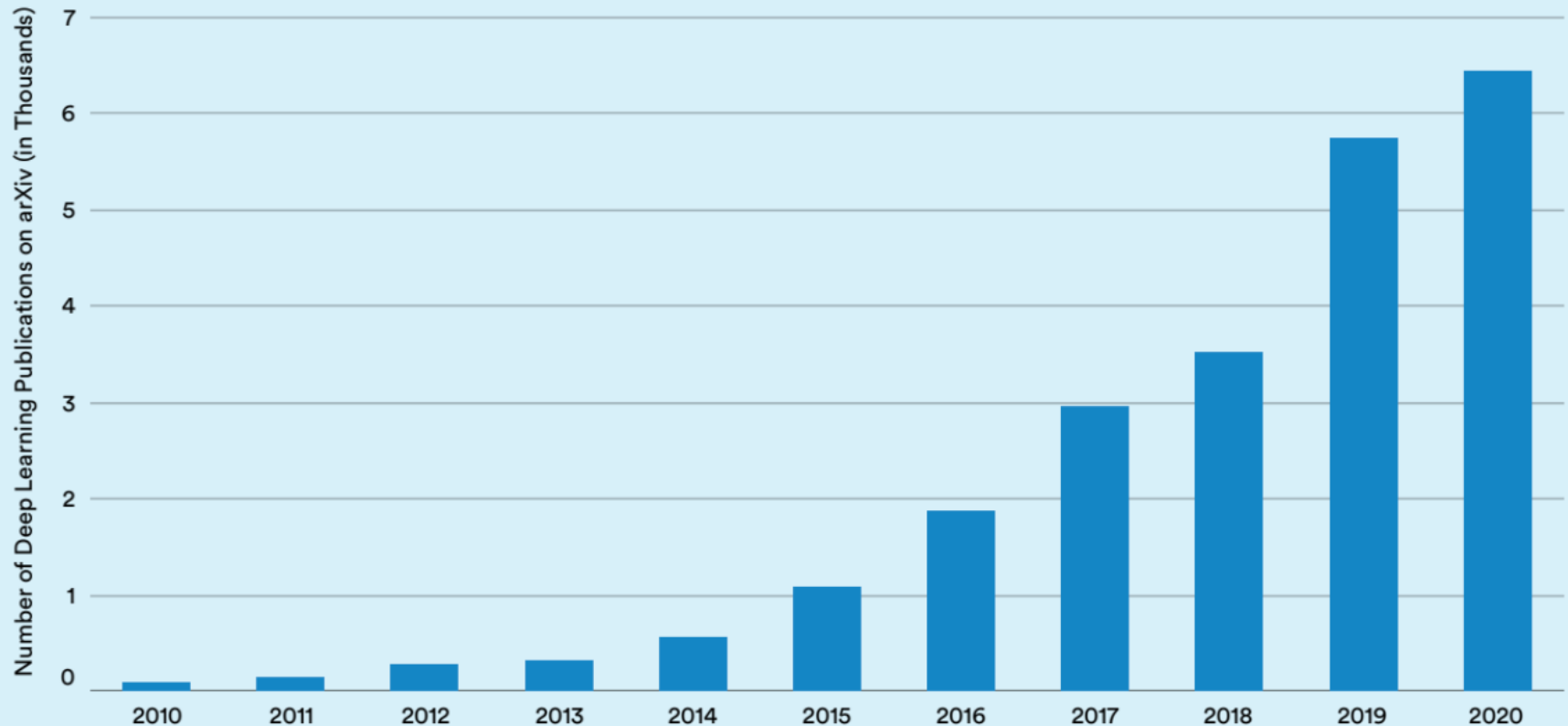


Interest in Deep Learning continues to increase – Research Community

<https://research.aimultiple.com/future-of-deep-learning/>

NUMBER of DEEP LEARNING PUBLICATIONS on ARXIV, 2010-19

Source: arXiv/Nesta, 2020 | Chart: 2021 AI Index Report

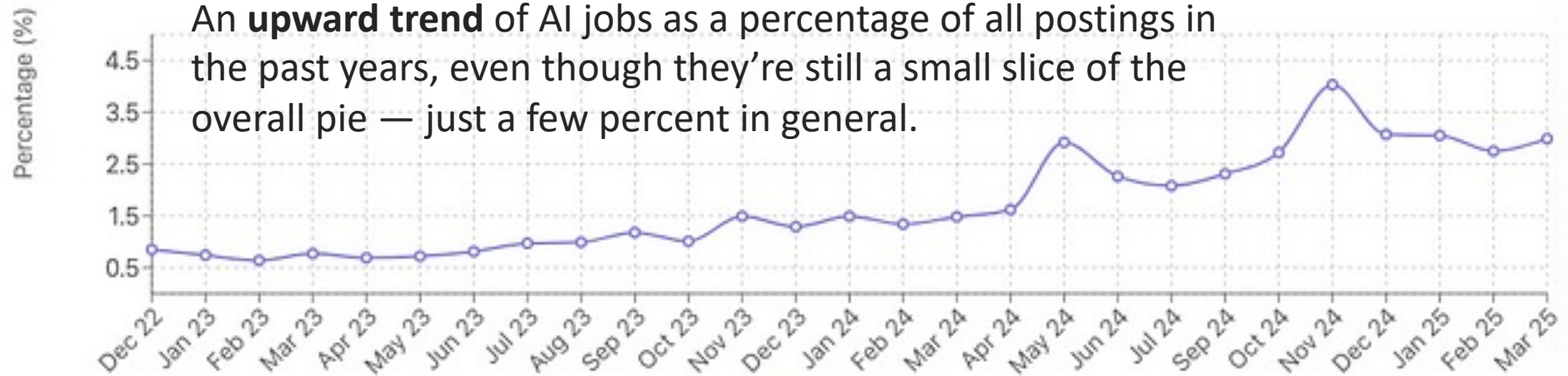


Deep Learning research effort continues to grow

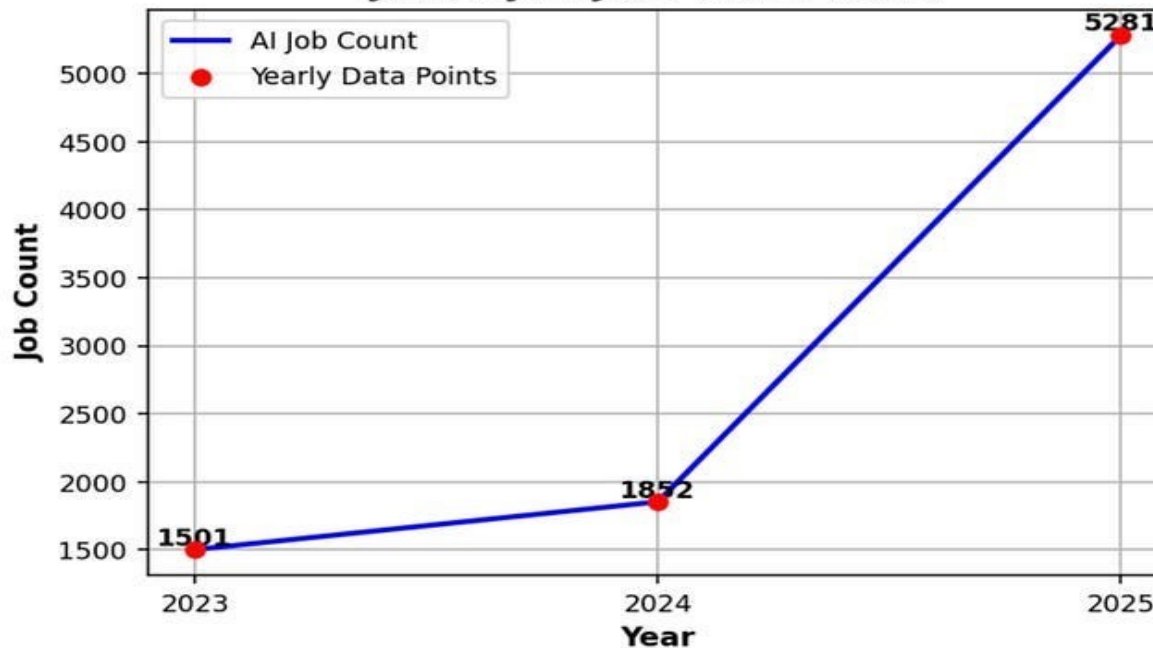
Are AI-related job postings increasing?

<https://medium.com/@vuthihienthu.ueb/should-data-scientists-pivot-to-ai-fb09d4836e8a>

Percentage of AI Jobs Out of All Data-Related Jobs



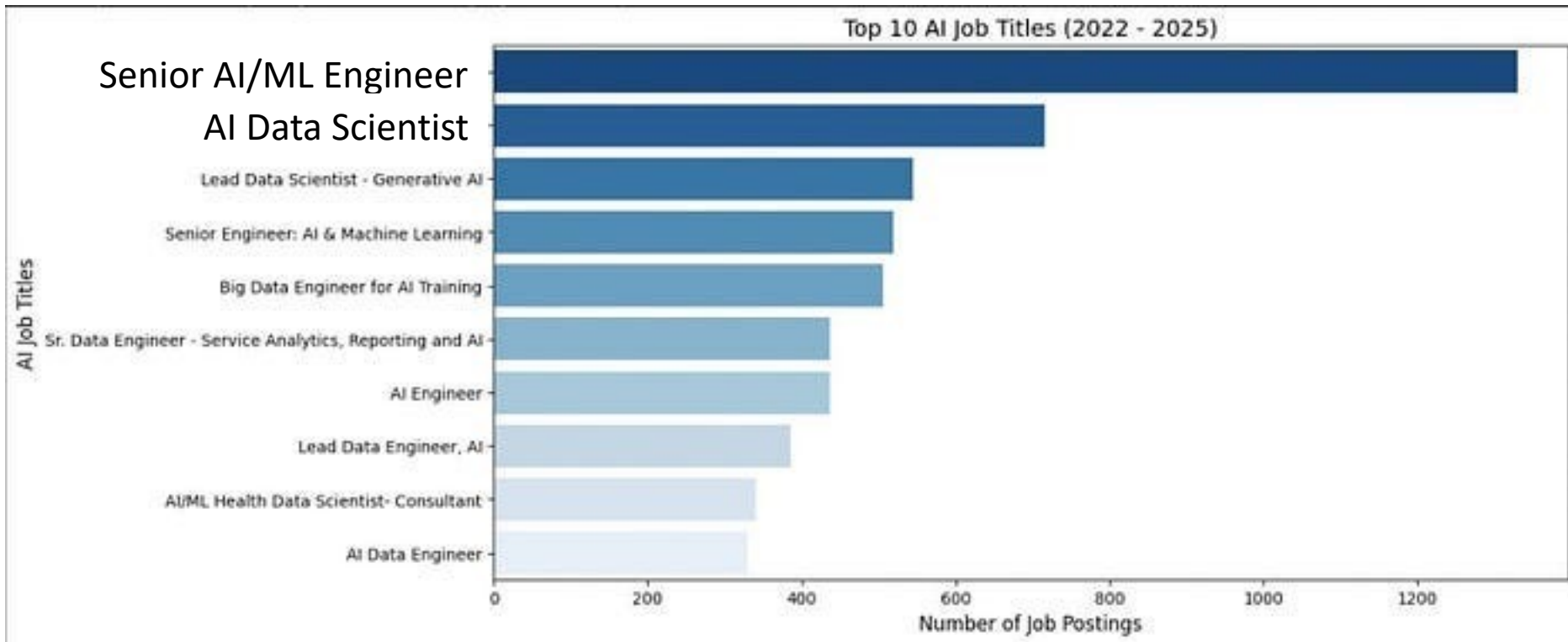
January AI Jobs (2023-2025)



A significant jump in AI job postings just one year.

About 5,000 in January 2025 compared to less than 2,000 in Jan 2024.

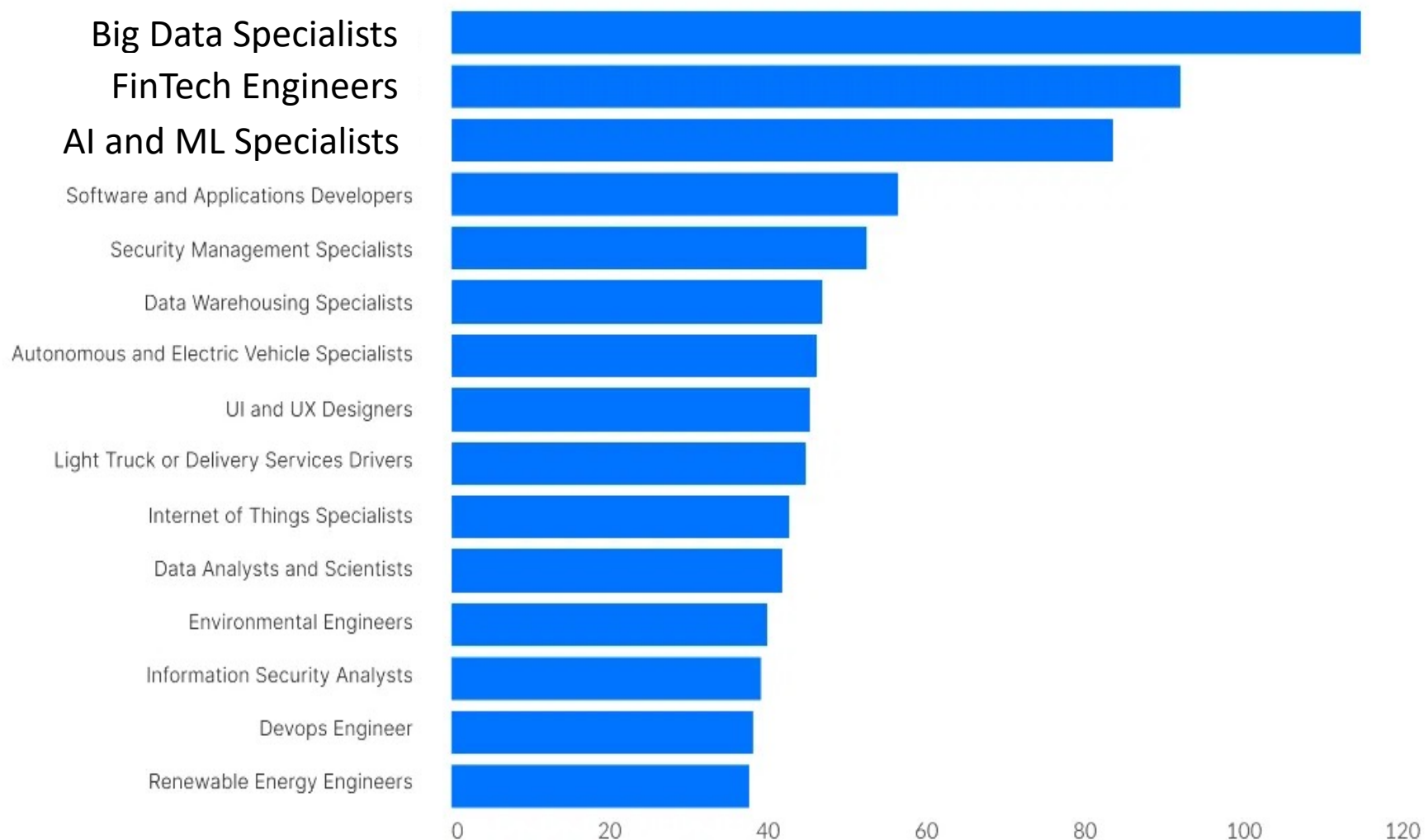
What Are These AI Jobs, Exactly?



What exactly does an AI Engineer do?

- AI Engineers lean towards the product and user end.
- They **don't BUILD** the AI models, but rather build applications that make use of pre-trained AI models to solve a specialized business problem.

Top fastest growing jobs, 2025-2030



Source: World Economic Forum, Future of Jobs Report 2025

What Are The Required Skills?

Three key layers of the AI engineering stack:

1. **Application Development** where you craft effective prompts, incorporating relevant context (like a knowledge base), potentially also building an agent workflow, and designing a solid user interface.
2. **Model Development** where you fine-tune models, optimizing inference, and handling data collection and engineering.
3. **Infrastructure layer** where you manage cloud resources and compute power, monitor and maintain AI applications at scale.

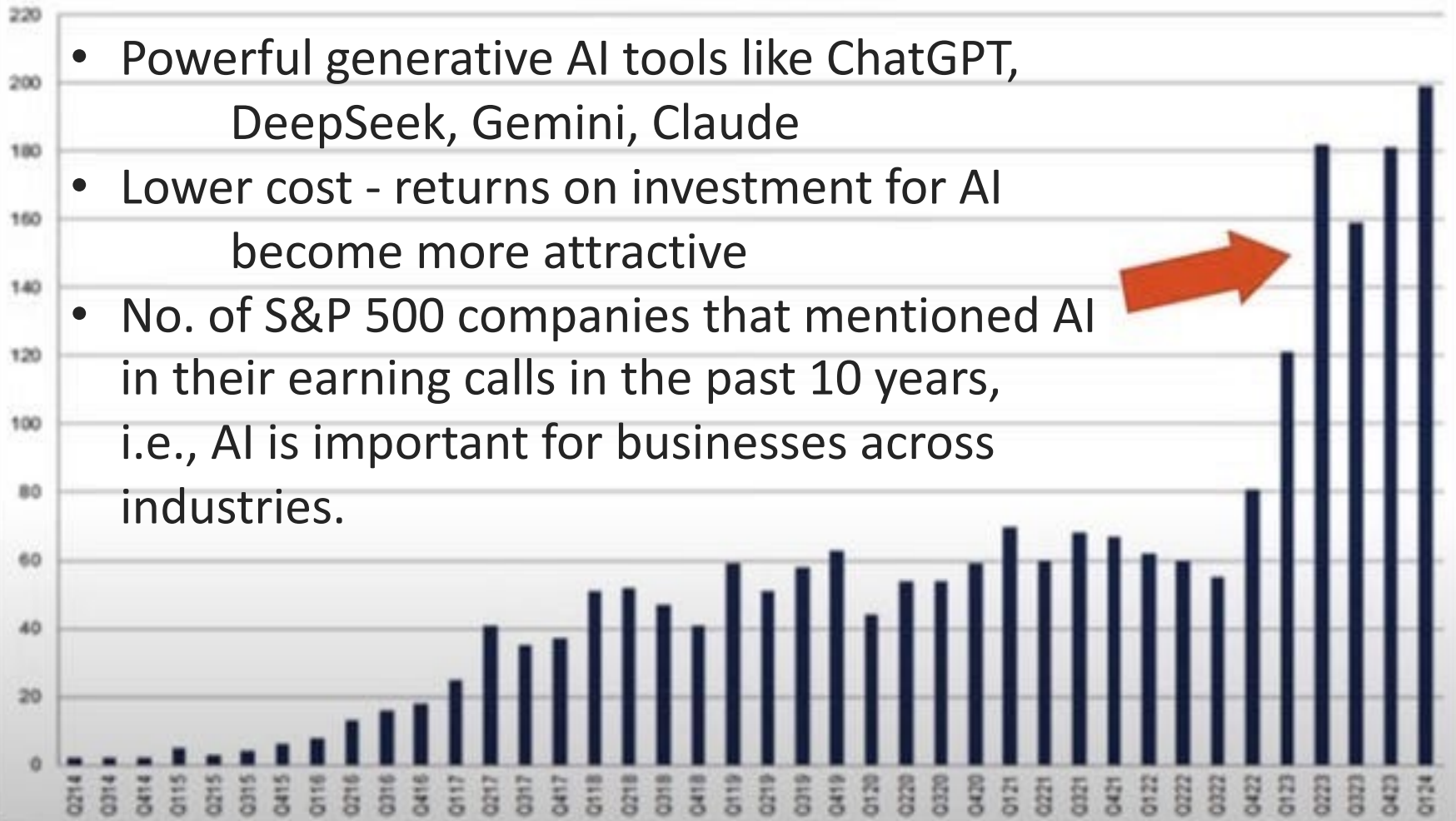
Skill/knowledge needed: Python, Pytorch, Azure, AWS

Course objectives: fundamentals and basic concepts of Deep Learning (not only on LLM for Application and Model development).

So... Why is This Happening?

of S&P 500 Cos. Citing "AI" on Earnings Calls: 10-Year
(Source: FactSet)

- Powerful generative AI tools like ChatGPT, DeepSeek, Gemini, Claude
- Lower cost - returns on investment for AI become more attractive
- No. of S&P 500 companies that mentioned AI in their earning calls in the past 10 years, i.e., AI is important for businesses across industries.



AI bubble

[CNBC News, August 18, 2025](#)

- OpenAI CEO Sam Altman has reportedly said that he believes **AI could be in a bubble**, comparing market conditions to those of the dotcom boom in the 1990s.
(OpenAI's annual recurring revenue is on track to pass \$20 billion this year, but that despite that, it remains unprofitable)
- AI the most important thing to happen in a very long time, but investors as a whole are overexcited about AI.
- Alibaba co-founder Joe Tsai, Bridgewater Associates' Ray Dalio and Apollo Global Management chief economist Torsten Slok have all raised similar warnings

5 Industries AI Will Completely Take Over by 2026

<https://pub.aimind.so/5-industries-ai-will-completely-takeover-by-2026-d16d36b70cce>

Goldman Sachs warns that AI could displace 300M jobs worldwide (not downsize/restructured, **replaced**), 76,440 jobs in 2025 alone.

- 1. Legal & Compliance** - *When Robots Know The Law Better Than Lawyers*
Reality: A lawyer can review 20 contracts/day. AI can process **2,000** with **99.7% accuracy**. (contract reviewer, paralegal researcher, compliance officer, document analyst, legal secretary)
- 2. Accounting & Finance** – *no coffee breaks, no calculation errors, work 24/7.*
Reality: JPMorgan's COIN (Contract Intelligence) program processes in seconds what used to take legal/accounting teams 360,000 hrs annually. (account payable/receivable, tax preparation/filing, audit procedure, financial reporting/analysis,, budget forecasting, expense management)
- 3. Healthcare Administrator** – *medical claims, reduce process time to minutes*
(medical billers, Insurance claim processors, Patient registration specialist, Medical records technician, authorization coordinator, data entry operator)
- 4. Customer Support & Call Centres** – *more patient and knowledgeable*
handle 80% of routine customer inquiries without human intervention
- 5. Content & Data Management** – write articles, analyse data, and generate reports faster (content writer, copywriter, market researcher, transcriptionists social media manager, administrative assistant)

DASC7606

Deep Learning

Teachers:

Dr. Bethany Chan 陳敏兒 (bethchan@hku.hk)

Professor Francis Chin 錢玉麟 (chin@hku.hk)

Course Learning Outcomes

- Able to **understand deep learning concepts**.
- Able to **understand the application** of **deep learning concepts** to various problems.

Topics Coverage

- **Perspectives**: history of AI, basic concepts, neural network, training
- **Vision**: CNN, AlexNet to Resnet, applications and concepts
- **Generative AI**: DeepDream, VAE, GAN
- **Natural Language Processing**: word embedding, RNN/LSTM, ELMo, Transformer, BERT, LLMs, GPT, ChatGPT, DeepSeek
- **Other topics**: Deep Reinforcement Learning, Ethics

Particulars about DASC7606 ...

Prerequisites

- Basic programming (Python)
- No prior knowledge in artificial intelligence or machine learning.
- Knowledge of **algorithms**, **calculus**, and **linear algebra** would be an advantage

Assessment

Assignments – 2 programming assignments (50%)

Examination – 2 hours (50%)

GPU farm and Python programming self-learn
(TA will give guidance to learn)

Others

Lectures will NOT be recorded, ppts posted after lecture

Breaks - 10 minutes + early end

Timetable

Lecture	Thursday (13:00-16:00)LE1; (19:00-22:00)CYP-P2	Tentative
1	September 4, 2025	
2	September 11, 2025	
3	September 18, 2025	Issue Assignment #1
4	September 25, 2025	
5	October 2, 2025	
6	October 9, 2025	Issue Assignment #2
	Reading Week	
7	October 23, 2025	Deadline for #1
8	October 30, 2025	
9	November 6, 2025	
10	November 13, 2025	
	November 20, 2025	Deadline for #2
Exam	December 3-23 (date unknown)	

Teaching Assistants

Solely responsible for the assignments

- DENG Weijian (Ken) kendwj@hku.hk
- CHEN Shu schen59@hku.hk

They will handle **ALL** matters
(supports, questions, submission, grading, ...)
related to assignments.

They can be contacted through Moodle, email or
any other means.